1 Worksheet

Question 1.1. Let $f : E \to [0, \infty]$ satisfy $\int_E f < \infty$. Let $A_n \subseteq A_{n-1} \subseteq E$ be a decreasing sequence of sets such that $m(A_n) \to 0$ as $n \to \infty$. Show that

$$\lim_{n\to\infty} \int_E f \mathbb{1}_{A_n} = 0$$

Proof. Let $\varepsilon > 0$ and let $\varphi : E \to [0, \infty]$ be a simple function satisfying $\varphi \le f$ and $(\int_E f) - \varepsilon < \int_E \varphi$. We then have the following chain of inequalities for any n:

$$0 \le \int_E (f - \varphi)\mathbb{1}_{A_n} \le \int_E (f - \varphi)\mathbb{1}_{A_n} + \int_E (f - \varphi)\mathbb{1}_{A_n^c} = \int_E (f - \varphi) < \varepsilon$$

Now we write $\varphi = \sum_{j=1}^k \alpha_j \mathbb{1}_{E_j}$ for some $E_j \subseteq E$ and get

$$\int_E \varphi \mathbb{1}_{A_n} = \int_E \sum_{j=1}^k \alpha_j \mathbb{1}_{E_j \cap A_n} \le \sum_{j=1}^k \alpha_j m(A_n)$$

which approaches 0 as $n \to \infty$. We then have

$$\int_E f \mathbb{1}_{A_n} < \int_E \varphi \mathbb{1}_{A_n} + \varepsilon \implies 0 \le \limsup_{n\to\infty} \int_E f \mathbb{1}_{A_n} \le \lim_{n\to\infty} \int_E \varphi \mathbb{1}_{A_n} + \varepsilon = \varepsilon$$

Since the above holds for all $\varepsilon > 0$, the limsup, hence the limit itself, of $\int_E f \mathbb{1}_{A_n}$ as $n \to \infty$ is 0. $\square$

Question 1.2. Let E be Lebesgue measurable with $m(E) > 0$. Show that there exists $p \in E$ so that $m(E \cap \{x : |x - p| < \varepsilon\}) > 0$ for all $\varepsilon > 0$. (Problem suggested by a student.)

Proof. Let $\varepsilon > 0$ and consider the sets $E_n := (n\varepsilon, (n+1)\varepsilon) \cap E$ and note that the union of all E_n for $n \in \mathbb{Z}$ is $E \setminus \{n\varepsilon : n \in \mathbb{Z}\}$ which has the same measure as E. We then have

$$m(E) = \sum_{n\in\mathbb{Z}} m(E_n)$$

and since $m(E) > 0$, we cannot have $m(E_n) = 0$ for all n. Thus for some $n \in \mathbb{Z}$, $m(E_n) > 0$, so we can take any $p \in E_n = (n\varepsilon, (n+1)\varepsilon) \cap E$ and we will have $(p - \varepsilon, p + \varepsilon) \supseteq (n\varepsilon, (n+1)\varepsilon)$, hence $m(E \cap (p - \varepsilon, p + \varepsilon)) \ge m(E_n) > 0$ as required. $\square$

Question 1.3. Let $f_n : \mathbb{R} \to [0, \infty]$ be a sequence of Lebesgue measurable functions that converge pointwise to some f. Assume that

$$\lim_{n\to\infty} \int_{\mathbb{R}} f_n = \int_{\mathbb{R}} f < \infty$$

Show that

$$\lim_{n\to\infty} \int_{\mathbb{R}} \mathbb{1}_E f_n = \int_{\mathbb{R}} \mathbb{1}_E f$$

for every measurable E. Show that this may fail if $\lim_{n\to\infty} \int_{\mathbb{R}} f_n = \int_{\mathbb{R}} f = \infty$.

Proof. Note that if we show that $\int_{\mathbb{R}} |f_n - f| \to 0$, then we will have

$$0 \le \left| \int_{\mathbb{R}} \mathbb{1}_E f_n - \int_{\mathbb{R}} \mathbb{1}_E f \right| \le \int_{\mathbb{R}} \mathbb{1}_E |f_n - f| \le \int_{\mathbb{R}} |f_n - f| \to 0$$

and hence the desired limit identity holds. For $\int_{\mathbb{R}} |f_n - f| \to 0$, we note that $f + f_n - |f_n - f|$ is either f or f_n for each x, hence nonnegative, and its liminf is $2f$. We thus have by Fatou's lemma

$$\begin{aligned}
2\int_{\mathbb{R}} f &= \int_{\mathbb{R}} \liminf_{n\to\infty} (f + f_n - |f_n - f|) \\
&\le \liminf_{n\to\infty} \int_{\mathbb{R}} (f + f_n - |f_n - f|) \\
&= \int_{\mathbb{R}} f + \liminf_{n\to\infty} \int_{\mathbb{R}} f_n - \limsup_{n\to\infty} \int_{\mathbb{R}} |f_n - f| \\
&= 2\int_{\mathbb{R}} f - \limsup_{n\to\infty} \int_{\mathbb{R}} |f_n - f| \\
\implies 0 \le \lim_{n\to\infty} \int_{\mathbb{R}} |f_n - f| &\le \limsup_{n\to\infty} \int_{\mathbb{R}} |f_n - f| \le 0
\end{aligned}$$

Thus we see that $\lim_{n\to\infty} \int_{\mathbb{R}} |f_n - f| = 0$, which completes the proof. $\square$

Question 1.4. Let $f_n : E \to [0, \infty]$ be a decreasing sequence of Lebesgue measurable functions such that $\int_E f_1 < \infty$. Let f be the pointwise limit. Show that

$$\lim_{n\to\infty} \int_E f_n = \int_E f$$

Proof. Just apply DCT with $g = f_1$. $\square$

Question 1.5. If $f : \mathbb{R} \to \overline{\mathbb{R}}$ is integrable, show that the function

$$F(x) := \int_{\mathbb{R}} \mathbb{1}_{(-\infty,x)} f$$

is continuous.

Proof. Let $x \in \mathbb{R}, \varepsilon > 0$, and consider the measurable sets $A_n = [x, x + 1/n)$, which is a decreasing sequence with $m(A_n) \to 0$. Note that $\int_{\mathbb{R}} |f| < \infty$ by integrability of f, so we can apply question 1 to get

$$\left| F\left(x + \frac{1}{n}\right) - F(x) \right| = \left| \int_{\mathbb{R}} \mathbb{1}_{[x,x+1/n)} f \right| \le \int_{\mathbb{R}} \mathbb{1}_{[x,x+1/n)} |f| \to 0$$

we can thus take $N > 0$ for which $\left| F\left(x + \frac{1}{N}\right) - F(x) \right| < \varepsilon$. It then follows that for all $h \in (0, 1/N)$, we have

$$|F(x+h) - F(x)| \le \int_{\mathbb{R}} \mathbb{1}_{[x,x+h)} |f| \le \int_{\mathbb{R}} \mathbb{1}_{[x,x+1/N)} |f| < \varepsilon$$

We can do the same for $A_n = (x - 1/n, x]$ and combine with the above to get a $\delta > 0$ for which $|F(y) - F(x)| < \varepsilon$ for all $y \in (x - \delta, x + \delta)$. Therefore, F is continuous. $\square$

2 Royden Chapter 4

Note that Royden uses the convention that a nonnegative measurable function is integrable when its Lebesgue integral is finite.

Question 2.1 (Royden 4.20)**.** Let $\{f_n\}$ be a sequence of nonnegative measurable functions that converges to f pointwise on E. Let $M \ge 0$ be such that

$$\int_E f_n \le M \quad \text{for all } n.$$

Show that $\int_E f \le M$. Verify that this property is equivalent to the statement of Fatou's Lemma.

Proof. Since we know Fatou's lemma to be true already, we will simply prove the equivalence of the property to it. If Fatou's lemma holds, then we must have

$$\int_E f \leq \liminf_{n\to\infty} \int_E f_n \leq M$$

as required. Conversely, suppose that whenever $f_n \to f$ pointwise on E with $\int_E f_n \leq M$ then $\int_E f \leq M$, and let $f_n \to f$ pointwise on E. If $\liminf \int_E f_n$ is infinite, then clearly the statement of Fatou's lemma holds. Otherwise, there is a subsequence f_{n_k} for which $\int_E f_{n_k}$ is a sequence approaching $\liminf \int_E f_n$. Since we still have $f_{n_k} \to f$ pointwise on E, for any $\varepsilon > 0$, we can take $K > 0$ such that for all $k \geq K$, we have $\int_E f_{n_k} < \liminf \int_E f_n + \varepsilon$, so by our property, we conclude

$$\int_E f \leq \liminf_{n\to\infty} \int_E f_n + \varepsilon$$

for all $\varepsilon > 0$. Taking $\varepsilon \to 0$, we prove Fatou's lemma where $f_n \to f$ pointwise everywhere on E. Since integrals over sets of measure zero are zero, this clearly extends to convergence a.e. $\square$

Question 2.2 (Royden 4.21)**.** Let the function f be nonnegative and integrable over E and $\epsilon > 0$. Show there is a simple function η on E that has finite support, $0 \leq \eta \leq f$ on E, and

$$\int_E |f - \eta| < \epsilon.$$

If E is a closed, bounded interval, show there is a step function h on E that has finite support and

$$\int_E |f - h| < \epsilon.$$

Proof. By definition of Lebesdue integral, we can take a simple function $\eta \leq f$ such that $\int_E f < \int_E \eta + \varepsilon/2$. This η might not have finite support, but we can use it to construct another simple function that has finite support and has integral close to one of η. Consider the increasing sequence $E_n := E \cap (-n, n)$ and simple functions $\eta_n := \eta \mathbb{1}_{E_n}$ with finite support. Since $\bigcup_{n=1}^{\infty} E_n = E$, we have $\eta_n \to \eta$ pointwise, and hence by MCT

$$\int_E \eta = \lim_{n\to\infty} \int_E \eta_n$$

Since the integral is finite, we can take n large enough for which $0 \leq \int_E \eta - \int_E \eta_n < \varepsilon/2$. By linearity of the integral, we have

$$\int_E (f - \eta_n) + \int_E \eta_n = \int_E f \Longrightarrow \int_E |f - \eta_n| = \left(\int_E f - \int_E \eta\right) + \left(\int_E \eta - \int_E \eta_n\right) < \varepsilon$$

as required.

If $E = [a, b]$ for $a < b$. In this case, any step function on E has finite support, so we don't need to worry about this condition. We first show that the η found above can be approximated by a step function:

Lemma 2.3. For any simple function φ and $\delta > 0$, we can find a step function $h : E \to [0, \infty]$ and a measurable $F \subseteq E$ such that $\varphi = h$ on F and $m(E \setminus F) < \delta$.

Proof. We note that a sum of finitely many step functions is also a step function, so it suffices to prove this for characteristic functions $\mathbb{1}_G$ with $G \subseteq E$ measurable. We take $O = \bigcup_{k=1}^{n} I_k$ where I_k are disjoint open intervals such that $m(O \setminus G) + m(G \setminus O) < \delta$ (the existence of such O can be proven by using outer regularity and the fact that open sets are countable unions of open intervals). We then define $h : E \to [0, \infty], x \mapsto \begin{cases} 1 & x \in O \\ 0 & x \notin O \end{cases}$, which is a step function because O is a disjoint union of open intervals. We then immediately see that $h(x) = 1 = \mathbb{1}_G(x)$ for $x \in G \cap O$ and $h(x) = 0 = \mathbb{1}_G(x)$ for $x \in E \setminus (O \cup G)$, so for $F = (G \cap O) \cup (E \setminus (O \cup G)) \subseteq E$, we have $h = \mathbb{1}_G$, and

$$\begin{aligned}
E \setminus F &= (E \setminus (G \cap O)) \cap (O \cup G) \\
&= ((E \setminus (G \cap O)) \cap O) \cup ((E \setminus (G \cap O)) \cap G) \\
&\subseteq ((G \cap O)^c \cap O) \cup ((G \cap O)^c \cap G) \\
&= (O \setminus G) \cup (G \setminus O) \\
\implies m(E \setminus F) &\leq m(O \setminus G) + m(G \setminus O) < \delta
\end{aligned}$$

as required. □

We take a simple η such that $\int_E |f - \eta| < \varepsilon/2$. Since η is simple, it has a finite bound $M > 0$, and we apply our lemma for $\delta = \frac{\varepsilon}{4M}$ to get a step function $h : E \to [0, \infty]$ and a set F such that $m(E \setminus F) < \delta$ and $\eta = h$ on F. By the construction in the lemma, we may assume that h is bounded above by M, so we have

$$\int_E |\eta - h| = \int_{E \setminus F} |\eta - h| \leq 2M \cdot m(E \setminus F) < \frac{\varepsilon}{2}$$

We then have

$$\int_E |f - h| \leq \int_E |f - \eta| + \int_E |\eta - h| < \varepsilon$$

□

Question 2.4 (Royden 4.25)**.** Let $\{f_n\}$ be a sequence of nonnegative measurable functions on E that converges pointwise on E to f. Suppose $f_n \leq f$ on E for each n. Show that

$$\lim_{n \to \infty} \int_E f_n = \int_E f.$$

Proof. □

Question 2.5 (Royden 4.28)**.** Let f be integrable over E and C a measurable subset of E. Show that

$$\int_C f = \int_E f \cdot \chi_C.$$

Proof. □